Newton Polygons and two Irreducibility Criteria over $\mathbb{Q}_p$

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Abstract

We first construct $\overline{\mathbb{Q}}_p$, the algebraic closure of $\mathbb{Q}_p$, and comment on some extensions of $\mathbb{Q}_p$. Then, we define Newton Polygons for polynomials and prove a result regarding them. Using this result, we prove two irreducibility criteria for polynomials over $\mathbb{Q}$ and over $\mathbb{Q}_p$.

1 Preliminaries

1.1 Basic Assumptions

We assume some basic familiarity with Ring Theory, Field Theory, and the p-adic numbers. A short writeup on the p-adic numbers follows. We seek to define the closure of the p-adic numbers, and work with polynomials over this field.

1.2 p-adic Numbers

We begin with the p-adic numbers, $\mathbb{Q}_p$. I will quickly define them using the idea of the completion of a metric space. I wish not to elaborate too much on the process of completion much, as it has already been covered in the course. Nevertheless, I will define the basic functions and sketch out the process of completion, which will help us in the understanding of $\mathbb{Q}_p$.

Let p be a prime natural number. We begin by considering the function defined as $v_p : \mathbb{Z} \to \mathbb{R}_{\geq 0}$ as $v_p(n) = k$ where $p^k \mid n$, but $p^{k+1} \nmid n$, for nonzero n.

An alternative way to write this down is by using the prime factorisation of n for nonzero n. Say $n = p^k p_1^{a_1} \cdots p_r^{a_r}$ where $p_1, \cdots, p_r$ are primes distinct from p, each a_i is a positive integer, and $k \geq 0$. This representation of n is unique due to the Fundamental Theorem of Arithmetic (of course, up to order and units in $\mathbb{Z}$). We then define $v_p(n) = k$, i.e. the exponent of p in the prime factorisation of n.

Of course, we have not seen what happens to 0, as $p^k \mid 0$ for all k. In this case alone, we define $v_p(0) = \infty$ for convenience. The function v_p is called the p-adic valuation of n. It represents the degree to which n is divisible by p. It satisfies the properties expected of a valuation:

- $v_p(ab) = v_p(a) + v_p(b)$
- $v_p(0) = \infty$
- $v_p(a+b) \geq \min\{v_p(a), v_p(b)\}$

As an example, $v_2(24) = 3, v_3(81) = 4, v_5(22) = 0$.

We then define another function on $\mathbb{Q}$ as $|\ \ |_p : \mathbb{Q} \to \mathbb{R}_{\geq 0}$ as:

$$\left|\frac{a}{b}\right|_p = \begin{cases} p^{v_p(b)-v_p(a)} & a \neq 0 \\ 0 & a = 0 \end{cases}$$

This is called the p-adic norm, and that it is a norm is given by the following three properties of $|\ \ |_p$:

- $|xy|_p = |x|_p \, |y|_p$
- $|x|_p = 0$ if and only if $x = 0$
- $|x+y|_p \leq |x|_p + |y|_p$. In fact, $|\ \ |_p$ satisfies a stronger version of this inequality due to the third property of v_p, which is that $|x+y|_p \leq \max\{|x|_p, |y|_p\}$. This is the ultrametric inequality, a stronger version of the triangle inequality.

We see that this norm induces a metric, and under this metric, $(\mathbb{Q}, |\ \ |_p)$ is not a Cauchy Complete space. Completing the space as one would expect gives us the field of p-adic numbers, $\mathbb{Q}_p$ which, with $|\ \ |_p$, is Cauchy complete. It is not Dedekind complete, as it cannot be an ordered field. Thus, $\mathbb{Q}_p$ is not isomorphic to $\mathbb{R}$, and in fact, for distinct primes p_1, p_2, the fields $\mathbb{Q}_{p_1}$ and $\mathbb{Q}_{p_2}$ are also not isomorphic to one another. Thus, these are indeed distinct field extensions/completions of $\mathbb{Q}$!

To say that these are *all* of the possible completions of $\mathbb{Q}$ as a metric space will require a bit of work, in the form of Ostrowski's Theorem, which states that the only possible nontrivial norms on $\mathbb{Q}$ are either equivalent (with respect to Cauchy Sequences) to the Euclidean norm (denoted $|\ \ |_\infty$) or to some p-adic norm.

One important property to note is that the image of v_p on the nonzero p-adic numbers is $v_p(\mathbb{Q}_p^\times) = \mathbb{Z}$. The valuation on $\mathbb{Q}_p$ can be defined either in terms of p-adic expansions of numbers, or by considering elements as limits of sequences, and assigning to each element, the limit of the valuations of the terms of the sequence that converges to it.

We also have the discrete valuation ring of v_p in $\mathbb{Q}_p$, denoted by $\mathbb{Z}_p$, the p-adic integers, consisting of all elements of non-negative p-adic valuations. This is, topologically, the closed ball around 0 of radius 1 in the the p-adic norm.

$$\mathbb{Z}_p := \left\{ x \in \mathbb{Q}_p \middle| \, |x|_p \leq 1 \right\} = \left\{ x \in \mathbb{Q}_p \middle| v_p(x) \geq 0 \right\}$$

I will not go too much into the structure of each of these, as they have been extensively discussed during the lectures. Results regarding these will be quoted as and when required.

1.3 Some properties of $\mathbb{Q}_p$ and $\mathbb{Z}_p$

One nice property about $\mathbb{Q}_p$ (as is true of $\mathbb{R}$) is that it is a locally compact field, i.e. given any point $x \in \mathbb{Q}_p$, there is a neighbourhood $B(x) \subseteq \mathbb{Q}_p$ such that $B(x)$ contains x and is sequentially compact (i.e. any sequence over $B(x)$ has a subsequence that converges in $B(x)$). In $\mathbb{R}$, this is quite easy to see, every element x is contained in some closed interval $[x-a, x+a]$. However, what is nice about these intervals is that they translate, so we can just find a compact neighbourhood of 0 and we are done. For this, the interval $[-1, 1]$ suffices, and we have that the interval $[x-1, x+1]$ is compact and contains x. In $\mathbb{Q}_p$, we follow a similar approach. We'll see that $\mathbb{Z}_p$ is a compact neighbourhood of 0 and that $x + \mathbb{Z}_p$ is a compact neighbourhood of x for each $x \in \mathbb{Q}_p$.

Proposition 1.1 : $\mathbb{Z}_p$ is compact in $\mathbb{Q}_p$.

Proof: A standard proof of this result would use the fact that $\mathbb{Z}_p \cong \lim_{n\to\infty} \mathbb{Z}/\langle p^n \rangle$, as topological groups, and as topological spaces. Thus, we can identify $\mathbb{Z}_p$ as a subspace of $\prod_{n=1}^{\infty} \mathbb{Z}/\langle p^n \rangle$, which is an intersection of closed sets of the product space. Tychonoff's Theorem then proves $\mathbb{Z}_p$ to be compact. However, I am yet to take a formal course in Topology, and may be misunderstanding some ideas in the proof. Consider this alternative, where we show that $\mathbb{Z}_p$ is sequentially compact in $\mathbb{Q}_p$.

Say $\{a_n\}_{n=1}^{\infty}$ is a sequence of p-adic integers. We are aware that we can write each a_n as

$$a_n = a_n^{(0)} + a_n^{(1)} p + a_n^{(2)} p^2 + \cdots, \quad a_n^{(i)} \in \{0, 1, \cdots, p-1\}$$

First consider the sequence $\{a_n^{(0)}\}_{n=1}^{\infty}$. This is a sequence over a finite set, and thus, at least one element residue $c_0 \in \{0, 1, \cdots, p-1\}$ that occurs infinitely many times in this sequence. Consider the subsequence $\{a_0, a_{n_1}, a_{n_2}, \cdots\}$ where each term $a_{n_k} \equiv c_0 \pmod{p}$. Now, all terms in this sequence (possibly with the exception of a_0) are elements of $c_0 + p\mathbb{Z}_p$. We repeat the same process by looking at the residues modulo p^2 as being of the form $c_0+c'p$ where c' is another residue modulo p. By restricting our collection to precisely those whose residues modulo p^2 occur an infinite number of times, we find another subsequence, $\{a_0, a_{n_1}, a_{n_{k_1}}, a_{n_{k_2}}, \cdots\}$, where every element (possibly with the exception of a_0, a_{n_1}) is in $c_+c_1p+p^2\mathbb{Z}_p$ (note that c_1 is the residue mod p that occurs infinitely many times in the second step). We repeat this process indefinitely and obtain, recursively a subsequence of terms, denoted, say, $\{x_0, x_1, x_2 \cdots\}$ which converges to the p-adic integer $c_0 + c_1p + c_2p^2 + \cdots$. Thus, we have a subsequence that converges in $\mathbb{Z}_p$, and thus, $\mathbb{Z}_p$ is sequentially compact. □

The following corollary is important, as we'd like to understand field extensions of $\mathbb{Q}_p$, leading up to the closure, $\overline{\mathbb{Q}}_p$, for which we'll prove results on vector spaces over locally compact fields.

Corollary 1.2 : $\mathbb{Q}_p$ is locally compact

Proof: For any $x \in \mathbb{Q}_p$, look at $x + \mathbb{Z}_p = \{y \in \mathbb{Q}_p \mid |y - x|_p \leq 1\}$. This is a compact neighbourhood of x as $\mathbb{Z}_p$ is compact. Thus, $\mathbb{Q}_p$ is locally compact. □

In the next section, we'll first look at some properties of extension fields of $\mathbb{Q}_p$ by looking at them as vector spaces over a locally compact field. Before we start the process, we state a few results regarding this that will be helpful:

Lemma 1.3 : Let V be a finite-dimensional vector space over a locally compact field, F, with a norm $| \ \ |_F$. The following hold:

- The sup-norm given by any basis of V *is* a norm on the vector space. Furthermore, V is locally compact.
- If $\| \ \ \|_V$ is a norm on V, the set $\{x \in V \mid \|x\|_V = 1\}$ is compact.
- All norms on V are equivalent
- When $V = K$ is a field extension of F, there is at most one norm on V that extends the norm on F.

2 The Algebraic Closure of $\mathbb{Q}_p$

Given any field, F, we can always find a field $\overline{F}$ such that $F \subseteq \overline{F}$ and $\overline{F}$ is algebraically closed. If we restrict this to saying that the extension $\overline{F}/F$ is algebraic (i.e. any element in $\overline{F}$ satisfies a polynomial over F), then, any two such fields are isomorphic to one another, so we can speak of this as a unique field, which we call the algebraic closure of F. For some standard examples, the closure of $\mathbb{R}$ is the field of complex numbers, $\mathbb{C}$, and the closure of $\mathbb{Q}$ is the field $\overline{\mathbb{Q}} = \mathbb{A}$, the field of Algebraic Numbers. The process of augmentation from $\mathbb{R}$ to $\mathbb{C}$ is simple, it is just a quadratic extension, whereas the latter is not so The degree of the extension $\overline{\mathbb{Q}}/\mathbb{Q}$ is infinite! Consider the following subfields of $\overline{\mathbb{Q}}$ as a chain: $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3}) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}) \subseteq \cdots$. At any stage, the newer field formed has degree 2 over the previous field, and thus, the n^{th} field has degree 2^n over $\mathbb{Q}$. The limit field is a subfield of the algebraic numbers which has infinite degree over $\mathbb{Q}$, and thus, $\overline{\mathbb{Q}}/\mathbb{Q}$ is not a finite extension. This is not a rare case, and is in fact, quite common. The closure of a field F is usually not a finite degree extension. The same holds for $\mathbb{Q}_p$.

As we saw earlier, the valuation of any nonzero element in $\mathbb{Q}_p$ is an integer. Consider the polynomial $f_{n,m}(X) = X^n - p^m$ where $n \nmid m$. Say this has a root in $\mathbb{Q}_p$, call it α. Then, $\alpha^n = p^m$. Applying the p-adic valuation, we see that $nv_p(\alpha) = mv_p(p) = m$. Thus, $v_p(\alpha) = \dfrac{m}{n} \notin \mathbb{Z}$. Thus, α can exist in an extension field of $\mathbb{Q}_p$, but not in $\mathbb{Q}_p$ itself. Thus, we must look at algebraic extensions of $\mathbb{Q}_p$.

2.1 Extension of norms

When we constructed the extension field $\mathbb{Q}_p$ of $\mathbb{Q}$, we did so by completing the metric space $(\mathbb{Q}, |\ \ |_p)$. However, on doing this, the metric/norm had to be extended to account for the new elements in $\mathbb{Q}_p$. When we consider algebraic extensions of $\mathbb{Q}_p$, we'll have to do something similar, and this section aims to discuss this (note that this process is similar to $\mathbb{Q} \to \overline{\mathbb{Q}} \to \mathbb{C}$ in the usual Archimedean sense).

We have already defined the norm with respect to a field extension. For this, let K/F be a simple field extension of finite degree, say n. For $\alpha \in K$ a primitive element, i.e. $K = F(\alpha)$, consider the multiplication map. When treating K as a vector space over F, this multiplication map is a linear operator on V, and it has a matrix, M_α with respect to a fixed basis. For this matrix, the trace and determinant are independent of the basis we selected, so, we may define a function $\mathbb{N}_{K/F}(\alpha) = \det M_\alpha$. This is the norm with respect to the field extension.

Seeing as K/F is of finite degree, it is algebraic. There are two equivalent definitions of this norm, one based on the minimal polynomial, and one based on its conjugates.

- Say $\min_{\alpha,F}(x) = x^n + a_1x^{n-1} + \cdots + a_n$ is the minimal polynomial of α over F. Then, $\mathbb{N}_{K/F}(\alpha) = (-1)^n a_n$

- Say $\tau_1, \cdots, \tau_n$ are all of the F-homomorphisms of K into a Galois extension of F containing K. Then, $\mathbb{N}_{K/F}(\alpha) = \tau_1(\alpha) \cdots \tau_n(\alpha)$, i.e. the product of all Galois conjugates of α in a splitting field of $\min_{\alpha,F}$

This definition extends to any $\beta \in F(\alpha)$ by using the determinant of the multiplication map. However, if we note that if $\beta \in F(\alpha)$, then, $F \subset F(\beta) \subset F(\alpha)$, so, by viewing $F(\beta)/F$ as well as $F(\alpha)/F(\beta)$ as vector spaces, we get corresponding bases, and taking products over these bases generates a basis of $F(\alpha)/F$ with respect to which the matrix of the multiplication map m_β has square blocks along the diagonal, and 0-blocks elsewhere. Thus, $\mathbb{N}_{K/F}(\beta) = \mathbb{N}_{F(\beta)/F}(\beta)^{[K:F(\beta)]}$. Furthermore, this norm is multiplicative, i.e. for any $a, b \in K$, we have that $\mathbb{N}_{K/F}(ab) = \mathbb{N}_{K/F}(a)\mathbb{N}_{K/F}(b)$.

As a final result, if $F \subset F(\beta) \subset K$ is a chain of field extensions of finite degree, then, $\mathbb{N}_{K/F}(\beta) = \mathbb{N}_{F(\beta)/F}(\beta)^{[K:F(\beta)]}$.

We now start looking at possible ways to extend the norm to algebraic extensions of $\mathbb{Q}_p$.

Say α is algebraic over $\mathbb{Q}_p$, and assume, for nontriviality, that it is not in $\mathbb{Q}_p$. Let $\min_\alpha(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ be its minimal polynomial over $\mathbb{Q}_p$. The field extension $\mathbb{Q}_p(\alpha)/\mathbb{Q}_p$ need not be Galois, so we consider a field $K \supset \mathbb{Q}_p(\alpha) \supsetneq \mathbb{Q}_p$ which is the splitting field of $\min_\alpha$. In this field, $\min_\alpha$ has distinct roots (as $\operatorname{char}(\mathbb{Q}_p) = 0$), call them $\alpha^{(1)}. \cdots, \alpha^{(n)}$ with $\alpha = \alpha^{(1)}$. Say $\| \ \ \|$ is a norm on K that extends $| \ \ |_p$.

By **Lemma 1.3**, there is at most one such norm. Thus, if $\| \ \ \|$ actually exists, it will be unique. If α' is a conjugate of α over $\mathbb{Q}_p$, then, there is a unique $\mathbb{Q}_p$-automorphism of K which sends $\alpha \mapsto \alpha'$. Call this σ. Now, let $\| \ \ \|'$ be defined as a norm on K such that $\|x\|' = \|\sigma(x)\|$. This is also a field norm on K and it extends $| \ \ |_p$. Thus, $\| \ \ \|' = \| \ \ \|$. Thus, $\left\|\alpha^{(i)}\right\| = \left\|\alpha^{(1)}\right\| = \|\alpha\|$ for all i. Thus,

$$\begin{aligned}
\left|\mathbb{N}_{\mathbb{Q}_p(\alpha)/\mathbb{Q}_p}(\alpha)\right|_p &= \left\|\mathbb{N}_{\mathbb{Q}_p(\alpha)/\mathbb{Q}_p}(\alpha)\right\| \quad (\text{as } \mathbb{N}_{\mathbb{Q}_p(\alpha)/\mathbb{Q}_p}(\alpha) \in \mathbb{Q}_p) \\
&= \left\|\prod_{i=1}^{n} \alpha^{(i)}\right\| \\
&= \prod_{i=1}^{n} \left\|\alpha^{(i)}\right\| \\
&= \|\alpha\|^n
\end{aligned}$$

Thus, we have that $\|\alpha\| = \left|\mathbb{N}_{\mathbb{Q}_p(\alpha)/\mathbb{Q}_p}(\alpha)\right|_p^{1/[\mathbb{Q}_p(\alpha):\mathbb{Q}_p]}$. Due to the earlier result, the expression on the right hand side is equal to $\left|\mathbb{N}_{L/\mathbb{Q}_p}(\alpha)\right|_p^{1/[L:\mathbb{Q}_p]}$ where L is any finite extension of $\mathbb{Q}_p(\alpha)$.

This will now be the extended p-adic norm. For convenience, we denote this $| \ \ |_p$ as well, with the obvious alteration to the domain, now being $\overline{\mathbb{Q}}_p$, as opposed to $\mathbb{Q}$. By definition, this takes up values in $p^{\mathbb{Q}} \cup \{0\}$. The issue is that we are yet to prove that this definition is indeed, a norm on the field.

The properties of being positive definite and multiplicative are obvious, however, the triangle inequality (we wish that the ultrametric inequality is still satisfied) is yet to be proven. It will follow from the following lemma.

Lemma 2.1 : Let $K/\mathbb{Q}_p$ be a finite degree field extension. Then, for all $\gamma \in K$, we have that if $|\gamma|_p \leq 1$, then, $|1+\gamma|_p \leq 1$.

Proof: It suffices to work with $K = \mathbb{Q}_p(\gamma) = \mathbb{Q}_p(1+\gamma)$. We then have

$$|\gamma|_p = \left|\mathbb{N}_{\mathbb{Q}_p(\gamma)/\mathbb{Q}_p}(\gamma)\right|_p^{1/[\mathbb{Q}_p(\gamma):\mathbb{Q}_p]}$$

$$|1+\gamma|_p = \left|\mathbb{N}_{\mathbb{Q}_p(\gamma)/\mathbb{Q}_p}(1+\gamma)\right|_p^{1/[\mathbb{Q}_p(\gamma):\mathbb{Q}_p]}$$

We fix $K = \mathbb{Q}_p(\gamma)$. As γ is primitive, the set $\Gamma = \{1, \gamma, \cdots, \gamma^{n-1}\}$ is a basis for $K/\mathbb{Q}_p$ as a vector space (n is the index $[\mathbb{Q}_p(\gamma) : \mathbb{Q}_p]$). With respect to this basis, we have the sup-norm, $\|\ \ \|$ on K.

We define a sup-norm (using the same notation) on the vector space $M_n(\mathbb{Q}_p)$ as $\|A\| = \max\left\{|A_{ij}|_p \,\middle|\, 1 \leq i,j \leq n\right\}$. We now work with the matrix $A = M_\gamma$, the matrix of the multiplication map for γ. We have that $A^k = M_{\gamma^k}$ for any $k \in \mathbb{N}$, and and $I + A = M_{1+\gamma}$. Thus, for any polynomial $P(X) \in \mathbb{Q}_p[X]$, we have that $P(A) = M_{P(\gamma)}$.

We claim that the sequence $\left\{\left\|A^i\right\|\right\}_{i=1}^\infty$ is bounded. If otherwise, we have a subsequence $\{\left\|A^{i_j}\right\|\}_{j=1}^\infty$ such that $\left\|A^{i_j}\right\| > j$ for each j. Let $b_j = \left\|A^{i_j}\right\| = |\beta_j|_p$, where β_j is an entry of A^{i_j} with maximal norm. Let $B_j = A^{i_j}/\beta_j$, so that $\|B_j\| = 1$ for each j. Let $\mathfrak{A} = \left\{X \in M_n(\mathbb{Q}_p) \,\middle|\, \|X\| = 1\right\}$.

As the vector space $M_n(\mathbb{Q}_p)$ is one over a locally compact field, the set $\mathfrak{A}$ is compact. Thus, as the B_j's are a sequence over $\mathfrak{A}$, there is a subsequence $\{B_{j_k}\}_{k=1}^\infty$ that converges to a matrix $B \in \mathfrak{A}$.

Now, $\det(B_j) = \det(A^{i_j}/\beta_j) = \det(A^{i_j})/\beta_j^n$. So,

$$\begin{aligned}\det(B_j) &< \det(A^{i_j})/j^n \\ &= \left|\mathbb{N}_{K/\mathbb{Q}_p}(\gamma)^{i_j}\right|_p /j^n \\ &= |\gamma|_p^{ni_j}/j^n \\ &\leq 1/j^n\end{aligned}$$

Thus, $\lim_{j\to\infty} \det B_j = 0$. Thus, $\det B = 0$ by the definition of convergence in the sup-norm (and due to continuity of det in the sup-norm). Thus, there is a nonzero vector v, written with respect to Γ such that $Bv = 0$. Clearly, $\Gamma' = \{l, \gamma l, \cdots, \gamma^{n-1} l\}$ is also a basis of K over $\mathbb{Q}_p$. If we show that $B\gamma^i v = 0$ for any i, we show that B is identically the zero matrix, and thus, $B \notin \mathfrak{A}$. Clearly, $B\gamma^i v = BA^i v = A^i B v = 0$ as $BA^i = A^i B$ due to B being a limit of scalar (over K) multiples of powers of A. We now have a contradiction. Thus the sequence $\{\|A^i\|\}_{i=1}^{\infty}$ is bounded in $\mathbb{R}$, say by $C > 0$.

We have, due to the ultrametric inequality, that $|\det(A)|_p \leq \|A\|^n$. Consider, for $N \in \mathbb{N}$, the matrix $(I+A)^N = I + NA + \binom{N}{2}A^2 + \cdots + NA^{N-1} + A^N$. Then,

$$\begin{aligned} |1+\gamma|_p^N &= \left|\det(I+A)^N\right|_p^{1/n} \\ &\leq \left\|(I+A)^N\right\| \\ &\leq \max_{1\leq i\leq N} \left\|\binom{N}{i} A^i\right\| \\ &\leq \max_{1\leq i\leq n} \left\|A^i\right\| \\ &\leq C \end{aligned}$$

Thus, $|1+\gamma|_p \leq C^{1/N}$. As N was arbitrary, we can take the limit as $N \to \infty$, and we get $|1+\gamma|_p \leq 1$. □

Corollary 2.2 : For all $\alpha, \beta \in K$, $|\alpha+\beta|_p \leq \max\{|\alpha|_p, |\beta|_p\}$.

Proof: This can easily be seen by making an assumption, without loss of generality, that $|\alpha|_p \geq |\beta|_p$, and then $|1+\beta/\alpha|_p \leq 1$ as $|\beta/\alpha|_p \leq 1$. The result follows from here.

Thus, $|\ \ |_p$ is truly a norm on the extension fields.

We now extend v_p as well. Note that on $\mathbb{Q}_p^\times$, we had the relation between v_p and $|\ \ |_p$ as $v_p(x) = -\log_p(|x|_p)$ where $\log_p$ is the usual base p logarithm. We do the same for $\overline{\mathbb{Q}}_p^\times$. Thus, on $\overline{\mathbb{Q}}_p$, the valuation of an element can be any rational number (or ∞ in the case of 0 alone).

2.2 Construction of $\overline{\mathbb{Q}}_p$

From the conclusions of the previous section, we have, for any finite degree extension of $\mathbb{Q}_p$, a unique extension of the original p-adic norm, which agrees even on towers of fields. As $\overline{\mathbb{Q}}_p$ is the union of all such extensions, the new p-adic norm extends uniquely to all of $\overline{\mathbb{Q}}_p$. Associated to any element is its norm given by the earlier definition.

Given a finite extension K of $\mathbb{Q}_p$, we can construct the integral closure of $\mathbb{Z}_p$ in K, which is the set of all elements in K that are integral over $\mathbb{Z}_p$, i.e. $\left\{\alpha \in K \middle| \min_{\alpha,\mathbb{Q}_p}(X) \in \mathbb{Z}_p[X]\right\}$. This will be denoted $\mathbb{Z}_K$ or $\mathcal{O}_K$. This is a ring with a unique maximal ideal, which I will denote $\mathfrak{m}$. This can be summarised in the following proposition.

Proposition 2.3 : For $K/\mathbb{Q}_p$ a finite degree extension, let $\mathbb{Z}_K$ denote the integral closure of $\mathbb{Z}_p$ in K. Then, $\mathbb{Z}_K = \{x \in K \mid |x|_p \leq 1\}$, and it is a ring. Furthermore, it has a unique maximal ideal $\mathfrak{m} = \{x \in K \mid |x|_p < 1\}$.

Proof: That $\mathbb{Z}_K$ is a ring is clear. To verify that the integral closure can be expressed as a valuation ring of v_p in K, we do the following:

Say α is in the integral closure of $\mathbb{Z}_p$ in K. We will show that $|\alpha|_p \leq 1$. Say $\min_\alpha(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0 \in \mathbb{Z}_p[X]$. Then, on substituting α,into this polynomial, we have $-\alpha^n = a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1}$. Applying the norm,

$$\begin{aligned} |\alpha^n|_p &= \left|a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1}\right|_p \\ &\leq \max_{0\leq i\leq n-1}\{|a_i|_p \left|\alpha^i\right|_p\} \\ &\leq \max_{0\leq i\leq n-1}\{\left|\alpha^i\right|_p\} \end{aligned}$$

However, we know that for a nonzero α, that $|\alpha|_p, \left|\alpha^2\right|_p, \cdots, \left|\alpha^k\right|_p, \cdots$ is either a decreasing sequence, a constant sequence (of 1's), or an increasing sequence, based on whether $|\alpha|_p < 1, |\alpha|_p = 1$, or $|\alpha|_p > 1$. The latter case would contradict the inequality. Thus, one of the the first two cases holds, and thus, $\alpha \in \{x \in K \mid |x|_p \leq 1\}$.

Conversely, if $\alpha \in \{x \in K \mid |x|_p \leq 1\}$, then, the same is true of all of its conjugates over $\mathbb{Q}_p$. The coefficients of the minimal polynomial are then just the elementary symmetric functions in the conjugates of α, and thus, by the ultrametric inequality, have norm upper bounded by 1. Thus, $\min_\alpha(X) \in \mathbb{Z}_p[X]$. Thus, $\mathbb{Z}_K = \{x \in K \mid |x|_p \leq 1\}$.

That $\mathfrak{m}$ is an ideal of $\mathbb{Z}_K$ is clear, and that it is maximal is also clear. Its uniqueness is due to the structure of maximal ideals of a DVR ($\mathbb{Z}_K$ is a DVR for $K/\mathbb{Q}_p$ a finite extension). □

Due to the maximality as an ideal of $\mathfrak{m}$ in $\mathbb{Z}_K$, we have that $\mathbb{Z}_K/\mathfrak{m}$ is a field. In fact, it is a finite extension of $\mathbb{F}_p$.

Proposition 2.4 : With $\mathbb{Z}_K, \mathfrak{m}, K$ as in the above proposition, the field $\mathbb{Z}_K/\mathfrak{m}$ is an extension field of $\mathbb{F}_p$ and has finite degree over $\mathbb{F}_p$.

Proof: $\mathfrak{m} \cap \mathbb{Z}_p = p\mathbb{Z}_p$ from the definition of $\mathfrak{m}$. Thus, $\mathbb{Z}_K/\mathfrak{m}$ is a field whose elements are additive cosets of the form $a + \mathfrak{m}$. If $a, b \in \mathbb{Z}_p$, then $a + \mathfrak{m} = b + \mathfrak{m}$ if and only if $a - b \in \mathfrak{m}$. However, $a - b$ is also a $p-$adic integer, and thus, $a - b \in p\mathbb{Z}_p$. Thus, if we consider the field $\mathbb{Z}_p/p\mathbb{Z}_p$, it has a natural inclusion into $\mathbb{Z}_K/\mathfrak{m}$ as $\psi : \mathbb{Z}_p/p\mathbb{Z}_p \hookrightarrow \mathbb{Z}_K/\mathfrak{m}, a + p\mathbb{Z}_p \mapsto a + \mathfrak{m}$.

However, $\mathbb{Z}_p/p\mathbb{Z}_p \cong \mathbb{F}_p$, and so, we have that $\mathbb{Z}_K/\mathfrak{m}$ is an extension field of $\mathbb{F}_p$. Now the claim is that $[\mathbb{Z}_K/\mathfrak{m} : \mathbb{F}_p] < \infty$, and in fact, more strictly, the index $[\mathbb{Z}_K/\mathfrak{m} : \mathbb{F}_p] \leq [K : \mathbb{Q}_p]$. Say $[K : \mathbb{Q}_p] = n$. Then, we will show that any choice of $n + 1$ cosets of $\mathfrak{m}$ will be linearly dependent over $\mathbb{F}_p$. This will prove the bound on $[\mathbb{Z}_K/\mathfrak{m} : \mathbb{F}_p]$.

Say $\overline{a_1}, \cdots, \overline{a_{n+1}}$ are cosets of $\mathfrak{m}$ defined as $\overline{a_i} = a_i + \mathfrak{m}$. As $a_1, \cdots, a_{n+1}$ are elements of K, they are linearly dependent over $\mathbb{Q}_p$, so there are $p-$adic numbers $b_1, \cdots, b_{n+1} \in \mathbb{Q}_p$ such that $a_1 b_1 + \cdots + a_{n+1} b_{n+1} = 0$. We can then multiply this equation through by a suitable power of p such that all b_i's are in $\mathbb{Z}_p$, but at least one is not in $p\mathbb{Z}_p$. The image of this map under the reduction $\tau : \mathbb{Z}_K \to \mathbb{Z}_K/\mathfrak{m}$ as $a \rightsquigarrow \overline{a}$ is also the 0 class. So, $\overline{a_1}\overline{b_1} + \cdots + \overline{a_{n+1}}\overline{b_{n+1}} = 0$, where $\overline{b_i}$ is the image of b_i in $\mathbb{Z}_p/p\mathbb{Z}_p$.

As at least one b_i is not in $p\mathbb{Z}_p$, it follows, that at least one $\overline{b_i}$ is not the 0 class, and thus, we have that the classes $\overline{a_1}, \cdots, \overline{a_{n+1}}$ are linearly dependent over $\mathbb{F}_p$. Thus, the extension field $\mathbb{Z}_K/\mathfrak{m}$ has finite degree over $\mathbb{F}_p$. $\square$

The field $\mathbb{Z}_K/\mathfrak{m}$ is called the residue field of the extension $K/\mathbb{Q}_p$. It is a field extension of $\mathbb{F}_p$ of some finite degree, f, and f is called the residue degree of K.

2.3 Ramified Extensions of $\mathbb{Q}_p$

We have already seen, via looking at the quotient ring $R = \mathbb{Z}_p[X]/\langle f(X)\rangle$ as to when it can be decomposed via Hensel's lemma and lifting of idempotents.

In particular, say $f(X) \in \mathbb{Z}_p[X]$. Then, if f is irreducible over $\mathbb{Z}_p$, then, the reduction $\overline{f} = f \pmod{p}$ has the form $\overline{f} = g^r$ where g is some polynomial in $\mathbb{F}_p[X]$ and $r \in \mathbb{N}$. In this case, $\mathbb{Z}_p[X]/\langle f\rangle$ is a domain, and its fraction field is $\mathbb{Q}_p[X]/\langle f\rangle$ (with $f(X)$ a prime element in $\mathbb{Z}_p[X]$ and in $\mathbb{Q}_p[X]$, as both are UFDs).

In this case, $R/\langle p\rangle \cong \mathbb{F}_p[X]/\langle g^r\rangle$, and it has a unique maximal ideal, which is the image of the ideal $\langle g\rangle$ of $\mathbb{F}_p[X]$ under the reduction map modulo g^r given by $\tau : \mathbb{F}_p[X] \to \mathbb{F}_p[X]/\langle g^r\rangle$ as $P(X) \mapsto P(X) + \langle g^r\rangle$.

Thus, R has a unique maximal ideal containing p, M. Then, M is generated by p, β where β is some element whose image modulo p is g. As g^r becomes 0 in $R/\langle p\rangle$, we must have that $M^r \subseteq pR$, and thus we have a sequence of ideals, ultimately giving us an element $\pi \in M \setminus M^2$ such that $pR = \pi^r R = M^r$. Thus, we can write $p = u\pi^r$ where u is a unit. This is a way we can see ramification occurring in field extensions.

An alternate way to look at ramification in a field extension is to look at the range of the extended valuation function. Let $K/\mathbb{Q}_p$ be field extension of degree $n < \infty$. Then, the image of v_p on $K^\times$ is contained in the ring $\frac{1}{n}\mathbb{Z}$. As v_p is a homomorphism, the image is a subgroup of the additive group of this ring, and thus, is of the form $\frac{1}{e}\mathbb{Z}$ for some $e \mid n$. This positive integer e is called the ramification index of K, and is equal to the r we saw earlier. Here, we can find an element π such that $v_p(\pi) = 1/e$. This means we can write $p = u\pi^e$ for some unit u. Such an element is called a uniformiser, and it generates the unique maximal ideal of the integral closure of $\mathbb{Z}_p$ in K. What follows is a standard result.

Proposition 2.5 : Let $K/\mathbb{Q}_p$ be an extension of degree $n < \infty$. Say the index of ramification is e. Let f be the residue field degree. Then, $ef = n$.

Proof: Admittedly, there is a proof of this result that uses the fact that $\mathbb{Z}_K$ is a rank n free module over $\mathbb{Z}_p$. However, I would not like to present this proof as I have not yet done a formal course with anything to do with module theory, and thus, may be misunderstanding some ideas in the proof. What follows is a slightly more elementary proof that I came up with using the idea of forming a basis for $K/\mathbb{Q}_p$. The proof itself inherently has the idea that we can use an intermediate field which is unramified over $\mathbb{Q}_p$.

Let $\{\alpha_1, \cdots, \alpha_f\}$ be a basis of the residue field $k_K = \mathbb{Z}_K/\mathfrak{m}$ over $\mathbb{F}_p$. Then, we can lift these elements to $\{a_1, \cdots, a_f\} \subset \mathbb{Z}_K$ such that on reduction modulo π they yield the α_i's. We know that $p = \pi^e u$ for some unit u. We look at the chain of ideals in $\mathbb{Z}_K$, namely

$$\mathbb{Z}_K \supset \pi\mathbb{Z}_K \supset \pi^2\mathbb{Z}_K \supset \cdots \supset \pi^e\mathbb{Z}_K = p\mathbb{Z}_K$$

Say we have a $\mathbb{Q}_p$-linear combination of the set $\{a_j\pi^i\}$ that equals 0 (here, i runs from 0 to $e-1$, and j runs from 1 to f). Let this be $\sum_{i,j} c_{ij}a_j\pi^i = 0$. We can choose our c_{ij}'s in $\mathbb{Z}_p$ by multiplying through by a large enough power of p. Modulo the uniformiser π, we get that $\sum_j \overline{c_{0j}a_j} = 0 \pmod{\pi}$. Here, $\overline{c_0 j} \in \mathbb{F}_p$, $\overline{a_j} = \alpha_j$. By the independence of the α_j's, we have that $c_{0j} = 0$ for each j. We then repeat the calculation modulo $\pi^2, \pi^3, \cdots, \pi^{e-1}$, we get that all c_{ij}'s are 0.

Thus, we have that the set $\mathfrak{B} = \{a_j\pi^i \mid 1 \le j \le f, 0 \le i \le e-1\}$ is linearly independent over $\mathbb{Q}_p$. If we show that it is a basis of $K/\mathbb{Q}_p$, then, as $|\mathfrak{B}| = ef$, we have that $ef = [K : \mathbb{Q}_p] = n$.

Say $z \in K$. Then, we know that for some $m \in \mathbb{N}$, $z\pi^m \in \mathbb{Z}_K$. Thus, if we show that $\mathbb{Z}_K \subseteq \operatorname{span}_{\mathbb{Q}_p}(\mathfrak{B})$, we'll have shown that $K \subseteq \operatorname{span}_{\mathbb{Q}_p}(\mathfrak{B})$. Say $y \in \mathbb{Z}_K$. As $\{\alpha_1, \cdots \alpha_f\}$ was a basis of k_K over $\mathbb{F}_p$, we'll have that on reduction, $y \pmod{\pi} = c_{01}\alpha_1 + \cdots + c_{0f}\alpha_f$, for some $c_{01}, \cdots, c_{0f} \in \mathbb{F}_p$. Thus, we may write that $y \equiv c_{01}a_1 + \cdots + c_{0f}a_f \pmod{\pi}$. Call the element on the right $\overline{y}_\pi$.

Now, set $y_1 = y - \overline{y}_\pi$. Then, we have that $y_1 \in \pi\mathbb{Z}_K$. We then reduce y_1 modulo π^2 to get $y_1 \pmod{\pi^2} = c_{11}\alpha_1 + \cdots + c_{1f}\alpha_f$. Thus, $y_1 \equiv c_{11}a_1 + \cdots + c_{1f}a_f \pmod{\pi^2}$. Call the element on the right $\overline{y}_\pi^2$. Then, set $y_2 = y - (\overline{y}_\pi + \pi\overline{y}_{\pi^2})$. We repeat the process until π^e. We then have:

$$y \equiv \sum_{i=0}^{e-1}\sum_{j=1}^{f} c_{ij}\pi^i a_j \pmod{p}, \quad \text{with } c_{ij}\text{'s in } \mathbb{F}_p$$

We repeat the process for higher powers of p (which correspond to the index m in the following summation) and eventually get:

$$y = \sum_{i=0}^{e-1} \pi^i \left(\sum_{j=1}^{f} a_j \left(\sum_{m=0}^{\infty} c_{ij}^{(m)} p^m \right) \right)$$

The innermost series here is one that converges in $\mathbb{Q}_p$. Thus, we can write each as some $x_{ij} \in \mathbb{Q}_p$. Then,

$$y = \sum_{i=0}^{e-1}\sum_{j=1}^{f} x_{ij} a_j \delta^i$$

Thus, $K \subseteq \operatorname{span}_{\mathbb{Q}_p}(\mathfrak{B})$. Thus, $ef = n$. $\square$

One small comment I'd like to make before moving on to future sections is that the above proof inherently contains the idea that for any K, a finite extension of $\mathbb{Q}_p$, there is an intermediate field K^{unr} which is unramified over $\mathbb{Q}_p$ (i.e. its ramification index is 1). This field is actually uniquely determined by the residue field degree of K. For each $f \in \mathbb{N}$, we have a unique field K_f^{unr} which is unramified over $\mathbb{Q}_p$, and of degree f. This is obtained by adjoining a primitive $(p^f - 1)^{\text{th}}$ root of unity to $\mathbb{Q}_p$.

I do not want to talk too much about Eisenstein Polynomials as a proof of their irreducibility is one of the final results of this project. However, if $K/\mathbb{Q}_p$ is a finite index with e, f as usual, then, $K = K_f^{\text{unr}}(\pi)$ where π is a root of an Eisenstein Polynomial of degree e. Thus, any finite extension of $\mathbb{Q}_p$ can be written as $\mathbb{Q}_p(\zeta_{p^f-1}, \pi_e)$.

3 Newton Polygons

In this section, I aim to define Newton Polygons for polynomials (working over $\mathbb{Q}_p$ or $\overline{\mathbb{Q}}_p$) and derive a relation between the Newton Polygon and the roots of the polynomial.

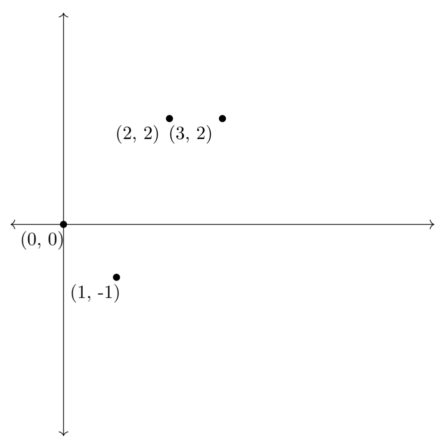
3.1 Newton Polygons for Polynomials

Say $f(X) \in \mathbb{Q}_p[X]$ is a polynomial. As we are working over a field, we may set $f(0) = 1$, as our goal will eventually be to derive an irreducibility criterion.

Let $f(X) = 1 + a_1 X + \cdots + a_n X^n$. We now make a plot in the plane (a usual visualisation of $\mathbb{R}^2$) using the following points:

$$N_f = \{(0,0), (1, v_p(a_1)), \cdots, (n, v_p(a_n))\}$$

For example, consider the polynomial $f(X) = 1 + \dfrac{1}{2}X + 4X^2 + 4X^3$ in the context of $\mathbb{Q}_2$. We plot the points as:



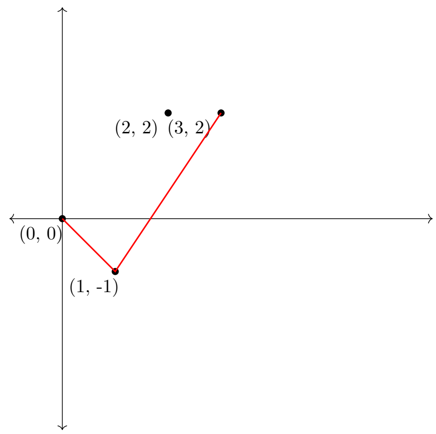
From here, we construct the lower convex hull of these plotted points. We do this via the following process:

- Begin with a vertical line passing through (0, 0). Rotate this line anti-clockwise until it hits a point in N_f.

- Once there is a collision, use the new point as a pivot, and rotate the line further. If there are multiple points hit simultaneously, the pivot is chosen to be the one with the largest x-coordinate.

- Repeat this process until $(n, v_p(a_n))$ is hit.

At each collision, we then draw the line segment defined by the earlier pivot and the new pivot. The resulting shape is the Newton Polygon of f. In the above example,



If at any instance when making a plot, we have a coefficient being 0, we treat the point as being plotted at infinity.

One remark to make is that these, although called Newton Polygons, are actually not polygons. However, their structures are simple, and their non-polygonal nature is not something that is of great importance.

Clearly, any given Newton Polygon is made up of line segments. To each segment, we associate its slope, λ, and its horizontal length, N (i.e. the difference in the x-coordinates of its endpoints). Due to the convexity of the Newton Polygons, the slopes of these segments increase from left to right.

We now prove a very important result, one that helps us identify the irreducibility criterion.

3.2 Roots from Newton Polygons

Theorem 3.1 : Let $f(X) = 1 + a_1X + \cdots + a_nX^n \in \mathbb{Q}_p[X]$. Say the Newton Polygon of f has a segment whose slope is λ and horizontal length is N. Then, counting multiplicities, there are precisely N roots of f in $\overline{\mathbb{Q}}_p$ whose valuations are all $-\lambda$.

Proof: Let $\alpha_1, \cdots, \alpha_n \in \overline{\mathbb{Q}}_p$ be the roots of f. Then, for each $1 \leq i \leq n$, let $\lambda_i = -v_p(\alpha_i)$.

We may order the α_i's such that $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$. Say the first r of these are equal, so $\lambda_1 = \cdots = \lambda_r < \lambda_{r+1}$. We now claim that the left-most segment of the Newton Polygon of $f(X)$ joins $(0, 0)$ to $(r, r\lambda_r)$.

Clearly, we have that the coefficients, $a_1, \cdots, a_n$ are elementary symmetric polynomials in the $1/\alpha_i$'s. Namely,

$$a_i = \pm \sum_{1 \le j_1 < \cdots < j_i \le n} \frac{1}{\alpha_{j_1} \cdots \alpha_{j_i}}$$

When we apply v_p, we get:

$$\begin{aligned} v_p(a_i) &\ge \min_{j_1 < \cdots < j_i} \{-v_p(\alpha_{j_1} \cdots \alpha_{j_i})\} \\ &= \min_{j_1 < \cdots < j_i} \left\{ -\sum_{k=1}^{i} v_p(\alpha_{j_k}) \right\} \\ &\ge i\lambda_r \end{aligned}$$

For $i = r$, the summations in the second-to-last line include exactly one of the form $-\sum_{k=1}^{r} v_p(\alpha_i) = r\lambda_r$. This is the minimum value any summation takes, and it is unique. Thus, by the Isosceles triangle principle applied to v_p, we have that $v_p(a_r) = r\lambda_r$.

For $i > r$, we have that $v_p(a_i) > i\lambda_r$. Thus, when plotting the Newton Polygon of $f(X)$, we see that there must be a change of slope at the point $(r, v_p(a_r))$. which I have just shown to be $(r, r\lambda_r)$. Thus, as no preceding point is a vertex (apart from $(0, 0)$), we must have that the first segment of the Newton Polygon joins $(0, 0)$ to $(r, r\lambda_r)$.

We can then repeat this proof for the next s of the λ_i's being equal, and can repeat this process until we have accounted for all (finitely many) λ_i's. Thus, the theorem is proven. $\square$

There is, in fact, an extension of this same theorem to power series over $\mathbb{Q}_p$, provided certain conditions on the power series. However, this is not our focus in this analysis.

The main takeaway from this theorem is that whenever we have a segment of the Newton Polygon of a polynomial, its length and slope are able to indicate properties of its roots in a closure. Of course, the earlier proof made no mention of whether we were working in $\mathbb{Q}_p$ or $\overline{\mathbb{Q}}_p$, so we could also take the theorem to hold true for polynomials over $\overline{\mathbb{Q}}_p$.

4 Irreducibility Criteria

We use the results of the previous section to prove some irreducibility criteria for polynomials.

Before we proceed, a definition follows. It will be clear why we study polynomials of the following kind.

Definition 4.1 (Pure Polynomials): Let $g(X) = 1 + a_1X + \cdots + a_nX^n$ be a polynomial over $\mathbb{Q}_p$ (or $\mathbb{Q}$). We say g is pure if the Newton Polygon of g has precisely one line segment joining $(0, 0)$ and $(n, v_p(a_n))$. If $g(X) \in \mathbb{Q}[X]$, and is pure when considered as an element of $\mathbb{Q}_p[X]$, we call it p-pure.

We note that a polynomial of the above form will be pure if and only if each point $(i, v_p(a_i))$ lies above the line joining $(0, 0)$ and $(n, v_p(a_n))$. As the line in question is defined by the equation $y = \frac{v_p(a_n)}{n}x$, we must have that $\frac{v_p(a_i)}{i} > \frac{v_p(a_n)}{n}$ for each $1 \leq i \leq n-1$. In all below discussions, the constant terms of polynomials are always nonzero.

Purity does not immediately imply irreducibility, nor does the converse hold. The polynomial $-p^2X^2+1$ is p-pure, but it factorises as $(1-pX)(1+pX)$. Conversely, X^2+1 is p-pure for every prime p, but is irreducible over $\mathbb{Q}_p$ whenever $p \not\equiv 1 \pmod 4$, as Hensel's Lemma gives us a p-adic squareroot of -1 precisely when $p \equiv 1 \pmod 4$.

However, when we adjoin certain other criteria to purity, we are able to derive some nice results on irreducibility.

4.1 Eisenstein Criterion

Let $f(x) = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_0$ be a polynomial over $\mathbb{Z}$. The Eisenstein Criterion states that such a polynomial is irreducible (over $\mathbb{Q}$) when:

- $\gcd(a_0, a_1, \cdots, a_n) = 1$
- There exists a prime p such that $p \mid a_i$ for each $0 \leq i < n$ and $p \nmid a_n$
- $p^2 \nmid a_0$

The polynomial is then said to be p-Eisenstein. For example, the polynomial $f(X) = X^4 + 12X^3 + 3X^2 + 27X + 15$ is 3-Eisenstein, and thus, irreducible. We have already covered a proof of this in the lectures. I would like to present one using Newton Polygons.

We translate the statement in terms of valuations. Naturally, the first condition on Eisenstein polynomials is easily achievable by taking a polynomial that satisfies the second and third criterion, and dividing through by the gcd of the coefficients (at least, in the broader sense, when working over $\mathbb{Q}$). We thus get a polynomial whose content is 1. Now, we say that $f(X) = a_0 + a_1 X + \cdots + a_n X^n$ is Eisenstein if there exists a prime p such that

- $v_p(a_i) \geq 1$ for each $0 < i < n$ and $v_p(a_n) = 0$
- $v_p(a_0) = 1$.

Criterion 1 (Eisenstein): Let $f(X) = a_0 + a_1 X + \cdots + a_n X^n$ be a polynomial over $\mathbb{Z}$ such that there exists a prime p with $v_p(a_i) \geq 1$ if and only if $i < n$ and $v_p(a_0) = 1$. Then, f is irreducible over $\mathbb{Q}_p$ (and thus, over $\mathbb{Q}$).

Proof: Say we are given such a polynomial $f(X)$ and consider a normalised version of this polynomial, $g(X) = f(X)/a_0$.

Now, we write $g(X) = 1 + b_1 X + \cdots + b_n X^n$, where $b_i = a_i/a_0$ for each i. Thus, $v_p(b_i) = v_p(a_i) - v_p(a_0) \geq 0$ for each $1 \leq i < n$. Furthermore, $v_p(b_n) = v_p(a_n) - v_p(a_0) = -1$.

As $v_p(b_i) \geq 0 > \dfrac{i v_p(b_n)}{n} = \dfrac{-i}{n}$ for each $1 \leq i < n$, we have that g is pure. Thus, the Newton Polygon of g has precisely one segment. The horizontal length of this segment is n, and its slope is $\dfrac{v_p(b_n)}{n} = \dfrac{-1}{n}$.

By the result obtained in the previous section, we have that there are precisely n roots of g in $\overline{\mathbb{Q}}_p$, say $\alpha_1, \cdots, \alpha_n$ (accounting for multiplicities) such that the valuation $v_p(\alpha_i) = \dfrac{1}{n}$ for each $1 \leq i \leq n$.

We can then write, over $\overline{\mathbb{Q}}_p$, that

$$g(X) = \left(1 - \frac{X}{\alpha_1}\right) \cdots \left(1 - \frac{X}{\alpha_n}\right)$$

Now, say $g(X) = Q(X)R(X)$ is a factorisation of g over $\mathbb{Q}_p$. As $\mathbb{Q}_p$ is a field, we can force $Q(0) = R(0) = 1$, i.e. that their constant terms are both 1. Say, without loss of generality, that $\deg(R) = r > 0$. Then, over $\overline{\mathbb{Q}}_p$, we can factorise $R(X)$ into linear factors. However, the roots of $R(X)$ will be precisely r of the roots of g. We may then re-order the α_i's such that $\alpha_1, \cdots, \alpha_r$ are the roots of

$R(X)$. We then have:

$$\begin{aligned} R(X) &= \left(1 - \frac{X}{\alpha_1}\right) \cdots \left(1 - \frac{X}{\alpha_r}\right) \\ &= 1 - s_1 X + \cdots + (-1)^r s_r X^r \end{aligned}$$

where the s_i's are the elementary symmetric polynomials in $\dfrac{1}{\alpha_1}, \cdots, \dfrac{1}{\alpha_r}$.

As we assumed $R(X) \in \mathbb{Q}_p[X]$, we must have that $s_r = \dfrac{1}{\alpha_1 \cdots \alpha_r} \in \mathbb{Q}_p$. Thus,

$$\begin{aligned} v_p(s_r) &= v_p\left(\frac{1}{\alpha_1 \cdots \alpha_r}\right) \\ &= -v_p(\alpha_1 \cdots \alpha_r) \\ &= -\sum_{i=1}^{r} v_p(\alpha_i) \\ &= \sum_{i=1}^{r} \frac{-1}{n} \\ &= \frac{-r}{n} \end{aligned}$$

As we need $s_r \in \mathbb{Q}_p^\times$, we must have that $v_p(s_r) \in \mathbb{Z}$. Thus, $n \mid r$. However, as $\deg(R) + \deg(Q) = \deg(g) = n$, we have that $r \leq n$. Thus, $r = n$ and $R(X) = g(X)$. Thus, g is irreducible over $\mathbb{Q}_p$.

AS $f(X) = a_0 \cdot g(X)$, we have that f is also irreducible over $\mathbb{Q}_p$. $\square$

Of course, the previous proof would work identically if we had considered polynomials over $\mathbb{Q}$, or even over $\mathbb{Q}_p$, in which case, we'd remove the condition of the existence of a prime, as p is a prime in $\mathbb{Z}_p$.

A neat generalisation to this criterion is one that I managed to propose and prove recently. In the above calculation, we would have made the same conclusion, i.e. that $n \mid r$ even if $v_p(s_r) = \dfrac{rk}{n}$ where $\gcd(k, n) = 1$.

We will see, once I prove the generalisation, that it actually implies the Eisenstein Criterion in a particular case, and is not equivalent to the Eisenstein Criterion. As the Eisenstein Criterion reduces the problem of irreducibility of an integer polynomial to the factorisation of n integers, so does the generalisation. However, the generalisation is for rational polynomials (which, of course, include integer polynomials as a subring!).

4.2 Dumas Criterion

Criterion 2 (Dumas): Let $f(X) = a_0 + a_1X + \cdots + a_nX^n$ be a polynomial over $\mathbb{Q}$. Say p is a prime such that the following hold:

- $a_0, a_n \neq 0$
- $\gcd(n, v_p(a_n - a_0)) = 1$
- For each $1 \leq i < n$, we have that $\dfrac{v_p(a_i) - v_p(a_0)}{i} > \dfrac{v_p(a_n) - v_p(a_0)}{n}$

Then, f is irreducible over $\mathbb{Q}_p$.

Before I prove this result, I would like to make a comment on the criterion. The first is that this can, of course, be generalised to polynomials over $\mathbb{Q}_p$. However, there is one more aspect to this which is nice, which is that this can be extended to irreducibility over a larger field, $\mathbb{Q}_p^{\mathrm{unr}}$, which is the union of all unramified extensions of $\mathbb{Q}_p$. The valuation ring of the extended p-adic norm in this field, $\mathbb{Z}_p^{\mathrm{unr}}$, is one where p is a prime as p does not ramify here.

Continuing the proof, we see that the proposition is very similar to the Eisenstein criterion. In fact, the proof for this is nearly identical.

Proof: Say $f(X)$ is such a polynomial, and let $g(X) = f(X)/a_0$. We write $g(X) = 1 + b_1X + \cdots + b_nX^n$.

As $g(X) \in \mathbb{Q}[X] \subset \mathbb{Q}_p[X]$, we have that g splits completely over $\overline{\mathbb{Q}}_p[X]$. It thus has, n roots, say $\alpha_1, \cdots, \alpha_n$, in $\overline{\mathbb{Q}}_p$, not necessarily distinct.

Clearly, $\dfrac{v_p(b_i)}{i} > \dfrac{v_p(b_n)}{n}$ for each $1 \leq i < n$ by simply applying the third condition and using the definition of b_i. Thus, g (and f) is pure. As such, the Newton Polygon of g consists of the single line segment joining $(0, 0)$ and $(n, v_p(b_n))$, which has slope $\dfrac{v_p(b_n)}{n}$.

Thus, $v_p(\alpha_i) = -\dfrac{v_p(b_n)}{n}$ for each $1 \leq i \leq n$. We can then write

$$g(X) = \left(1 - \frac{X}{\alpha_1}\right) \cdots \left(1 - \frac{X}{\alpha_n}\right)$$

Say $g(X) = Q(X)R(X)$. Then, say that $\deg(R) = r > 0$. Then, R has roots in $\overline{\mathbb{Q}}_p$, which are precisely r of the roots of $g(X)$. Thus, up to a reordering, $\alpha_1, \cdots, \alpha_r$ are the roots of R in $\overline{\mathbb{Q}}_p$. By forcing $Q(0) = R(0) = 1$, which is justified as we are working over a field, we get that

$$\begin{aligned} R(X) &= \left(1 - \frac{X}{\alpha_1}\right) \cdots \left(1 - \frac{X}{\alpha_r}\right) \\ &= 1 - s_1 X + \cdots + (-1)^r s_r X^r \end{aligned}$$

where the s_i's are the elementary symmetric polynomials in $\frac{1}{\alpha_1}, \cdots, \frac{1}{\alpha_r}$. Then, we know that $s_r = \frac{1}{\alpha_1 \cdots \alpha_r}$. Thus,

$$\begin{aligned} v_p(s_r) &= v_p\left(\frac{1}{\alpha_1 \cdots \alpha_r}\right) \\ &= -v_p(\alpha_1 \cdots \alpha_r) \\ &= -\sum_{i=1}^{r} v_p(\alpha_i) \\ &= \sum_{i=1}^{r} \frac{-v_p(b_n)}{n} \\ &= \frac{-r[v_p(a_n) - v_p(a_0)]}{n} \end{aligned}$$

Given that $\gcd(n, [v_p(a_n) - v_p(a_0)]) = 1$ and that $s_r \in \mathbb{Q}_p^\times$, we know that $v_p(s_r) \in \mathbb{Z}$. Thus, $n \mid r$, giving us that $n = r$. Thus, $R(X) = g(X)$ and g is irreducible. Consequently, f is irreducible over $\mathbb{Q}_p$. As all we needed was that $v_p(s_r) \in \mathbb{Z}$, we may even claim that f is irreducible over $\mathbb{Q}_p^{\mathrm{unr}}$. $\square$

That this criterion implies the Eisenstein Criterion is easy to see. As long as we set $v_p(a_n) = 0, v_p(a_0) = 1, v_p(a_i) \geq 0$ for each $1 \leq i < n$ in $f(X)$, we obtain that $g(X) = f(X)/a_0$ is a p-Eisenstein polynomial.

Furthermore, the Eisenstein criterion fixes the value of $v_p(a_n) - v_p(a_0)$ at -1, whereas the generalisation allows this to take any value coprime to n.

5 References

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3. Notes on Hensel's Lemma provided by the instructor on Moodle

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